

A Rule Utilitarian Analysis of Cybersecurity Disclosure in *Zero Days*

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Introduction

In this case study, I am going to describe how the rule utilitarianism normative ethical theory can be used to resolve an ethical challenge that Eric Chien faces in the film *Zero Days*. As cybersecurity companies process an increasingly substantial number of threats, employees begin to face more conflicting obligations. For people like Eric Chien, who was a member of Symantec's security response team, he knows that it's his professional responsibility to protect the public by disclosing threats that are perceived as having the greatest potential impact. However, he is simultaneously aware of the possible consequences of exposing what he even acknowledged as a likely state-level cyberweapon, as he became concerned for his own safety at the time. Using rule utilitarianism, I will argue that Chien did the morally right thing by deciding to publish the *W32.Stuxnet Dossier*, which supported the general moral rule of responsibly disclosing information that would maximize the total good, if all cybersecurity researchers did the same thing in such a scenario. Throughout this work, I will describe the main characteristics of rule utilitarianism, explain Chien's ethical challenge, and lastly demonstrate how the rule utilitarianism theory resolves the dilemma.

Description of Rule Utilitarianism

Rule utilitarianism focuses on acting in a particular way that adheres to a moral rule, unlike act utilitarianism, which influences an individual's decisions based on what produces the greatest increase in good for each affected individual in a particular scenario. Therefore, rule utilitarianism is more consistent, because any action taken while adhering to it will align with a morally correct rule. Both of these utilitarian ethical theories are similar in how both focus on the consequences of an action, as they are both consequentialist theories, which focus "on the

consequences of an action” (Quinn, 2025, p. 71). However, act utilitarianism has been described as affirming that “an action is good if its net effect” results in “more happiness than unhappiness” (Quinn, 2025, p. 71). In contrast, rule utilitarianism instructs individuals to consider whether an action is morally right based on whether its application consistently results “in the greatest net increase in happiness” (Quinn, 2025, p. 75). By having everyone adhere to a moral rule all the time, rule utilitarianism supports the idea of justifying actions that consistently increase the long-term welfare of all affected individuals.

Overview of the Situation

In this case study, I have decided to analyze Eric Chien’s actions in *Zero Days*. First, Chien was part of Symantec’s security response team at the time that the *Stuxnet* cyberweapon was discovered. He and colleague Liam O’Murchu became aware of *Stuxnet* when it was shared as a binary sample from a customer’s machine in Iran, and it was then that they assigned its name based on the combination of the strings “STUB” and “xnet” in its code. Eric noted that the code was unusually complex, as it took them weeks to analyze it and even begin to understand its payload, or its whole purpose. Eventually, they discovered multiple zero-day exploits and stolen digital certificates, and that *Stuxnet* specifically targeted computers running the Microsoft Windows operating system, but only those that used specific make/model combinations of hardware/software for Siemens-branded industrial control systems. This combination of discoveries made Eric and Liam realize that this appeared to be the work of a nation-state actor. Understandably, they became anxious of this cyberweapon that had physical, real-world impacts. As both of them were separately interviewed in the film, both of them admitted that they were fearful of being watched or listened to, especially once they realized the geopolitical implications

of *Stuxnet*. This resulted in Chien having to make a difficult decision about whether he should publish Symantec's findings.

Relevance of Rule Utilitarianism

As rule utilitarianism is a theory that focuses on consistently following moral rules that favor the long-term welfare of society, it can be used to determine if Chien acted in a way that maximizes the overall good for the global cybersecurity field. For example, by applying this ethical theory to his situation, we can consider how cybersecurity depends on consistent principles, such as various technical standards and protocols, along with professionalism. This emphasizes the strength of rule utilitarianism in this context, especially when compared to the utilitarian calculus that is applied to each moral decision in act utilitarianism. After all, imagine if every cybersecurity analyst either disclosed or hid information based on political pressure or other personal concerns. Aside from emphasizing consistency, it's important to connect rule utilitarianism to a specific rule that maximizes the overall good if everyone adopted it. For example, Chien's actions could be described as aligning with the rule that cybersecurity professionals should responsibly disclose vulnerabilities that endanger the public's safety. This differs from how he could've changed his decision based on another possible option he could've chosen. For example, imagine if he decided that cybersecurity researchers should hide their findings if the disclosure of the findings could threaten his personal safety or international stability. By adhering to the framework that the rule utilitarianism theory provides, it can be inferred that this ethical theory can be used to sufficiently evaluate Chien's actions.

Resolution of the Ethical Challenge

Eric Chien's ethical challenge focused on whether he should publicly disclose Symantec's findings about the *Stuxnet* malware. He knew that it would likely improve global cybersecurity, especially by promoting transparency. However, he was also aware that by disclosing the company's findings (especially by having his name attached to the report), he was risking political fallout, further escalations by another country (before the malware's target became more widely known), and his personal safety.

By relying on rule utilitarianism to determine what would make a moral rule correct, the theory can be used to consider the possible rules that could be used to guide Chien's actions. The first option would be to decide that cybersecurity professionals should always disclose vulnerabilities for public safety. To assess each possible rule, we should also consider both the short-term and long-term consequences of the action. For Option 1, we know that the potential short-term consequences include possibly revealing sensitive information or attracting political retaliation. However, regarding long-term consequences, Chien knew that by following this rule, he would be maintaining transparency that would improve global security, trust in the overall cybersecurity community, and defend public safety. Therefore, if everyone followed this rule, there are multiple implications that suggest that Option 1 could maximize long-term good. These include ensuring that less vulnerabilities are hidden from the public's knowledge, and that computer systems will be patched more efficiently.

Another option would be to decide that cybersecurity professionals should hide their findings about a vulnerability if its disclosure could result in a controversy or other personal risk. Short-term benefits for Option 2 include protecting personal safety and avoiding geopolitical tension. However, by keeping *Stuxnet* hidden from the public, multiple negative long-term

consequences would follow. These include weakening trust, allowing governments to take actions without being held accountable, and causing civilians to be unprotected from attacks. Therefore, if everyone followed the rule proposed by Option 2, secrecy would be favored, while the overall harm would be increased.

Accordingly, Option 1 would result in the greatest total good, even though it inevitably carried some personal risk. This would morally justify Chien's decision to publish the *W32.Stuxnet Dossier*, as it served as a contribution towards increasing the public's safety and trust in the long-term.

Conclusion

In summary, the ethical challenge that Chien is facing in *Zero Days* can be resolved by analyzing it with the rule utilitarianism ethics theory. From this perspective, I think that by choosing to disclose the findings that he and O'Murchu gathered from analyzing *Stuxnet*, Chien followed the morally correct rule, as it resulted in him taking an action that would consistently result in the largest increase in the overall good of society. Because rule utilitarianism is intended to be applied with a long-term outlook, Chien's decision to remain committed to responsibly disclose his findings aligns with a rule that would improve cybersecurity efforts worldwide, if the rule were followed by all cybersecurity researchers. His actions have demonstrated how consistent communication and transparency in cybersecurity studies can protect the public's safety and improve people's trust in computer infrastructure. The rule utilitarianism theory provides a unique approach to decision-making by not only focusing on the consequences of an action, but also by determining which moral rule would result in the largest increase in the overall good, if everyone in Eric Chien's situation followed the same rule.

References

Quinn, M. J. (2025). *Ethics for the information age* (9th ed.). Pearson Education.